

STORLIEN-STORVALLEN, Sweden

WMO: 022070

Lat: 63.283N

Lon: 12.122E

Elev: 583

StdP: 94.51

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-21.0	-27.0	0.3	-24.5	-23.2	0.5	-20.5	18.3	-4.7	15.5	-2.1	0.9	120	0.764

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	23.0	14.4	21.0	13.8	18.9	12.7	15.9	20.8	14.8	19.3	13.6	17.7	3.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	10.6	17.7	12.7	9.8	16.4	11.6	9.1	15.5	46.5	20.8	43.2	19.3	39.9	17.7	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.8	10.5	8.9	-27.9	25.6	4.0	2.2	-30.8	27.2	-33.1	28.5	-35.3	29.7	-38.2	31.3		
			-28.3	17.7	3.9	1.8	-31.1	19.0	-33.4	20.1	-35.5	21.1	-38.3	22.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-6.2	-6.2	-4.0	-0.1	4.6	8.9	12.1	11.2	7.6	2.2	-2.8	-5.2
	DBStd	8.03	5.84	6.03	4.79	3.70	4.02	3.80	3.83	3.40	3.32	3.97	5.27	6.19
	HDD10.0	3168	501	453	435	304	177	66	22	26	87	243	384	471
	HDD18.3	6011	759	686	693	554	427	284	197	223	323	501	634	729
	CDD10.0	202	0	0	0	0	8	32	86	62	13	1	0	0
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0
	CDH23.3	52	0	0	0	0	2	4	43	3	0	0	0	0
	CDH26.7	6	0	0	0	0	0	0	6	0	0	0	0	0
Wind	WSAvg	3.6	4.3	4.3	4.5	3.4	3.2	3.1	2.6	2.8	3.3	3.7	3.6	4.0
Precipitation	PrecAvg	885	77	64	61	42	55	85	103	97	90	75	67	69
	PrecMax	1141	160	152	127	100	93	127	186	165	199	117	164	146
	PrecMin	599	10	4	20	15	28	28	36	35	26	33	4	15
	PrecStd	110	41	31	32	23	16	25	38	36	48	23	46	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	4.8	5.8	11.9	22.3	23.6	27.5	23.6	19.4	14.2	8.6	5.4
		MCWB	1.8	2.3	2.7	6.2	12.2	13.6	16.5	14.7	12.2	9.5	6.4	3.9
	2%	DB	2.7	3.0	4.1	9.0	17.9	20.5	24.1	21.3	16.6	11.1	6.5	4.0
		MCWB	1.3	1.5	2.1	4.6	10.5	12.7	15.6	13.9	11.5	7.9	4.8	2.8
	5%	DB	1.8	1.8	3.1	6.8	14.3	18.1	22.0	19.4	14.8	9.4	4.7	3.0
		MCWB	0.7	0.5	1.5	3.4	8.8	11.4	14.8	13.2	10.6	6.9	3.1	1.7
	10%	DB	0.7	0.8	2.0	5.2	11.3	16.0	19.7	17.5	13.0	7.9	3.3	1.7
		MCWB	-0.4	-0.3	0.6	2.6	7.1	10.6	13.5	12.7	9.8	5.7	1.9	0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.1	3.6	6.8	13.1	15.7	18.1	16.2	13.6	10.4	7.0	4.3
		MCDB	3.3	4.1	4.3	10.6	20.3	20.4	25.0	20.5	16.9	12.7	8.5	4.9
	2%	WB	1.6	1.9	2.5	4.9	11.0	13.6	16.7	15.1	12.2	8.3	5.0	3.0
		MCDB	2.4	2.8	3.7	7.9	16.9	17.7	22.4	19.6	15.4	10.5	6.2	3.8
	5%	WB	0.8	0.9	1.6	3.8	9.2	12.2	15.5	14.0	11.1	7.1	3.3	1.8
		MCDB	1.7	1.6	2.9	6.3	14.1	16.9	20.4	17.9	14.2	9.1	4.5	2.8
	10%	WB	-0.2	-0.2	0.7	2.8	7.4	11.0	14.3	13.1	9.9	5.9	2.0	0.7
		MCDB	0.6	0.6	1.9	4.9	11.0	15.5	18.9	16.5	12.6	7.7	3.2	1.6
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.7	6.1	6.8	7.4	8.0	9.3	8.6	7.2	5.3	5.2	5.5
		MCDBR	4.8	4.8	5.8	8.9	13.8	12.8	14.4	12.1	10.5	7.2	5.5	5.1
		MCWBR	4.3	4.2	4.4	5.7	7.6	6.6	7.3	6.3	6.3	4.8	4.6	4.7
	5% WB	MCDBR	4.6	4.7	5.4	7.7	13.2	11.1	12.9	9.9	9.6	6.6	5.4	4.8
		MCWBR	4.2	4.5	4.4	5.0	7.6	6.1	6.8	5.7	6.1	4.8	4.7	4.7
Clear-Sky Solar Irradiance	taub	0.197	0.259	0.288	0.316	0.327	0.326	0.335	0.324	0.300	0.280	0.207	0.182	
	taud	2.012	2.216	2.232	2.268	2.411	2.492	2.481	2.505	2.558	2.449	2.073	1.954	
	Ebn at Noon	444	689	811	865	889	897	876	857	814	676	420	231	
	Edh at Noon	30	59	87	105	100	94	93	83	64	47	27	14	
All-Sky Solar Radiation	RadAvg	0.17	0.72	2.06	3.89	4.63	4.86	4.67	3.56	2.14	0.93	0.26	0.08	
	RadStd	0.02	0.10	0.24	0.30	0.32	0.45	0.52	0.37	0.21	0.11	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	-142	-178	N/A	N/A

Nomenclature: See separate page